

Section 11.1: Introduction

Goal:

So far, we have assumed $Y_1, \dots, Y_n$ are i.i.d.

- Thus,

This assumption is unrealistic for many inference problems! Some examples:

1. Mean stopping distance for a particular car can depend on

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2. Salary of new college graduates can depend on

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In this chapter, we'll have a random variable Y , the _____ variable.

- $E(Y)$ will be a function of _____.
- The $x_1, \dots, x_k$ are called _____.

Many different types of functions can be used to model a response that's a function of one or more independent variables. These can be classified into two categories:

1. deterministic model:

- e.g.,

2. probabilistic model:

- e.g.,

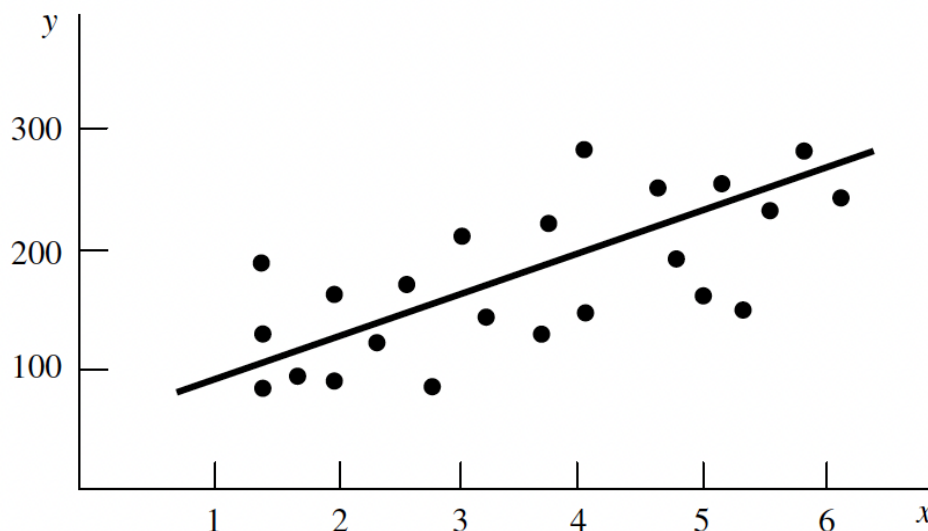
- β_0 and β_1 are

- Y and ϵ are

- * We assume that

- * Thus,

Consider Figure 11.1 from your textbook:



Section 11.2: Linear Statistical Models

Def. A **linear statistical model** relating a random response Y to a set of independent variables $x_1, \dots, x_k$ is of the form

where $\beta_0, \beta_1, \dots, \beta_k$ are unknown parameters, ϵ is a random variable, and $x_1, \dots, x_k$ assume known values. We will assume that $E(\epsilon) = \underline{\hspace{2cm}}$ and hence that

Remarks:

- If the model relates $E(Y)$ as a linear function of β_0 and β_1 only, the model is called a _____.
- If multiple independent variables—say $x_1, \dots, x_k$ —are of interest and we model $E(Y)$ by $E(Y) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$, the model is called a _____ linear regression (MLR) model.
- When we say we have a linear statistical model for Y , we mean that $E(Y)$ is a linear function of the unknown parameters $\beta_0, \beta_1, \dots, \beta_k$ and *not* necessarily a linear function of $x_1, \dots, x_k$.

Example 1. Determine if the following are linear statistical models.

(a) $E(Y) = \beta_0 + \beta_1 x + \beta_2 x^2$

(b) $E(Y) = \beta_0 + \beta_1 \cdot \ln(x)$

(c) $E(Y) = \beta_0 + e^{\beta_1}$

Note: The subsequent sections will focus on the simple linear regression model.

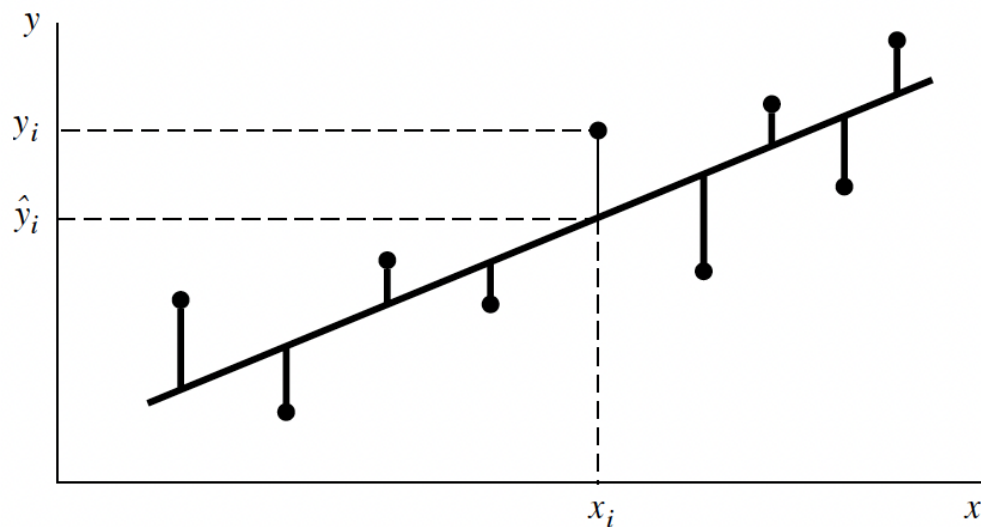
Section 11.3: The Method of Least Squares

We postulate that $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, $i = 1, \dots, n$, where $E(\epsilon_i) = 0$. The parameters β_0 and β_1 are unknown.

Goal:

Question. How will we achieve our goal?

Consider Figure 11.5 from your textbook:



We will minimize SSE using calculus.

Summary:**Least-Squares Estimators for the Simple Linear Regression Model**

1. $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$, where $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ and $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$.
2. $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.

Section 11.4: Properties of the Least-Squares Estimators**Goal:****Model:**

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n,$$

where $\epsilon_1, \dots, \epsilon_n$ are i.i.d. with mean 0 and variance σ^2 .

Properties of the Least-Squares Estimators; Simple Linear Regression

1. The estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased—that is, $E(\hat{\beta}_i) = \beta_i$, for $i = 0, 1$.
2. $V(\hat{\beta}_0) = c_{00}\sigma^2$, where $c_{00} = \sum x_i^2 / (nS_{xx})$.
3. $V(\hat{\beta}_1) = c_{11}\sigma^2$, where $c_{11} = \frac{1}{S_{xx}}$.
4. $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = c_{01}\sigma^2$, where $c_{01} = \frac{-\bar{x}}{S_{xx}}$.
5. An unbiased estimator of σ^2 is $S^2 = \text{SSE}/(n - 2)$, where $\text{SSE} = S_{yy} - \hat{\beta}_1 S_{xy}$ and $S_{yy} = \sum (y_i - \bar{y})^2$.

If, in addition, the ϵ_i , for $i = 1, 2, \dots, n$ are normally distributed,

6. Both $\hat{\beta}_0$ and $\hat{\beta}_1$ are normally distributed.
7. The random variable $\frac{(n - 2)S^2}{\sigma^2}$ has a χ^2 distribution with $n - 2$ df.
8. The statistic S^2 is independent of both $\hat{\beta}_0$ and $\hat{\beta}_1$.

Section 11.5: Inferences Concerning the Parameters β_i **Test of Hypothesis for β_i**

$$H_0 : \beta_i = \beta_{i0}.$$

$$H_a : \begin{cases} \beta_i > \beta_{i0} & \text{(upper-tail rejection region),} \\ \beta_i < \beta_{i0} & \text{(lower-tail rejection region),} \\ \beta_i \neq \beta_{i0} & \text{(two-tailed rejection region).} \end{cases}$$

$$\text{Test statistic: } T = \frac{\hat{\beta}_i - \beta_{i0}}{S\sqrt{c_{ii}}}.$$

$$\text{Rejection region: } \begin{cases} T > t_\alpha & \text{(upper-tail alternative),} \\ T < -t_\alpha & \text{(lower-tail alternative),} \\ |T| > t_{\alpha/2} & \text{(two-tailed alternative),} \end{cases}$$

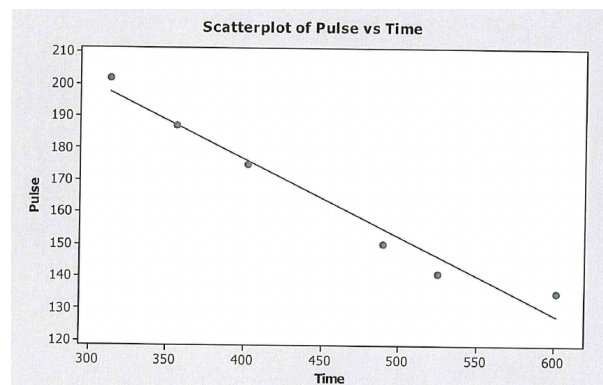
where

$$c_{00} = \frac{\sum x_i^2}{nS_{xx}} \quad \text{and} \quad c_{11} = \frac{1}{S_{xx}}.$$

Notice that t_α is based on $(n - 2)$ df.

Example 2. Six runners ran one mile. They then immediately recorded their finishing time (seconds), as well as their pulse rate (beats per minute). The data and corresponding scatterplot are given below. Test if finishing time is a useful predictor of pulse rate. Use $\alpha = 0.05$.

Time (seconds)	Pulse (bpm)
313	202
356	187
402	175
490	150
525	141
602	135



$100(1 - \alpha)\%$ Confidence Interval for β_i :

Example 3. Calculate a 95% confidence interval for β_1 from Example 2.